

Section 1.1

1.1.8

Write the number in the form $a + bi$:

$$\frac{(8 + 2i) - (1 - i)}{(2 + i)^2}$$

Solution.

□

1.1.10

Write the number in the form $a + bi$:

$$\left[\frac{2 + i}{6i - (1 - 2i)} \right]^2$$

Solution.

□

1.1.14

Show that $\operatorname{Re}(iz) = -\operatorname{Im} z$ for every complex number z .

Proof.

□

1.1.15

Let k be an integer. Show that

$$i^{4k} = 1, \quad i^{4k+1} = i, \quad i^{4k+2} = -1, \quad i^{4k+3} = -i.$$

Proof.

□

1.1.20

Solve each of the following equations for z .

1.1.20 (c)

$$(2 - i)z + 8z^2 = 0$$

Solution.

□

1.1.22

Find all solutions to the equation $z^4 - 16 = 0$.

Solution.

□

1.1.24

Let z be a complex number such that $\operatorname{Im} z > 0$. Prove that $\operatorname{Im}(1/z) < 0$.

Proof.

□

Section 1.2

1.2.7

Describe the set of points z in the complex plane that satisfies each of the following.

1.2.7 (d)

$$|z - 1| = |z + i|$$

Solution.

□

1.2.7 (e)

$$|z| = \operatorname{Re} z + 2$$

Solution.

□

1.2.7 (f)

$$|z - 1| + |z + 1| = 7$$

Solution.

□

1.2.16

Prove that if $|z| = 1$ ($z \neq 1$), then

$$\operatorname{Re} \left[\frac{1}{1 - z} \right] = \frac{1}{2}.$$

Proof.

□

Section 1.3

1.3.7

Find the argument of each of the following complex numbers and write each in polar form.

1.3.7 (d)

$$-2\sqrt{3} - 2i$$

Solution.

□

1.3.7 (g)

$$\frac{-1 + \sqrt{3}i}{2 + 2i}$$

Solution.

□

1.3.10

Show the following:

1.3.10 (a)

$$\arg(z_1 z_2 z_3) = \arg z_1 + \arg z_2 + \arg z_3$$

Proof.

□

1.3.10 (b)

$$\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$$

Proof.

□

1.3.12

Find the following.

1.3.12 (b)

$$\operatorname{Arg}(-\pi)$$

Solution.

□

1.3.12 (c)

$$\operatorname{Arg}(10i)$$

Solution.

□